

of lithium-ion battery started during the oil crisis of 1970s. Prof. M. Stanley Whittingham (USA) was working towards developing methods that could lead to fossil fuel-free energy technologies supported with research on superconductors. He discovered an extremely energy rich material from titanium-disulphide to create an innovative cathode in a lithium battery. This material, at molecular level, has spaces that can store intercalated lithium ions.

Anode of the battery was made partially from metallic-lithium having strong drive to release electrons. With present batteries offering potential much less than a volt even, such a combination of anode and cathode materials resulted in a battery with a great potential (just over two volts). However, the developed metallic lithium battery was reactive and too explosive to be viable.

Contemporary to this research, Prof. John B. Goodenough (USA) predicted that cathode made up of a metal-oxide instead of a metal-sulphide would have even greater potential. After a long systematic research, Prof. Goodenough in 1980 demonstrated that cobalt-oxide with intercalated lithium ions can produce potential as much as four volts. This was an important breakthrough in the history, leading to development of much more powerful batteries.

Making the research on cathode by Prof. Goodenough as the basis, Prof. Akira Yoshino (Japan) created the first commercially viable lithium-ion battery in 1985. Instead of using reactive lithium in the anode, he made use of petroleum coke (a carbon material) that like cathode of cobalt-oxide can intercalate lithium ions. Use of petroleum coke cathode resulted in a lightweight and hardwearing battery that could be charged hundreds of times before its performance deteriorated, thus leading to development of rechargeable batteries.

Lithium-ion batteries are not based upon chemical reactions, leading to the breakdown of electrodes; rather, these work on the basis of lithium ions flowing back and forth between anode and cathode. Since entry of lithium-ion batteries to the market in 1991, these have revolutionised every aspect

of our lives based on the use of batteries and have laid down the foundation of a wireless, fossil fuel-free society.

The Royal Swedish Academy of Sciences awarded the Nobel Prize in Chemistry 2019 to Profs John B. Goodenough (USA), M. Stanley Whittingham (USA) and Akira Yoshino (Japan) for the development of lithium-ion batteries. These lightweight, rechargeable and powerful batteries are now used in everything from cellphones to laptops and electric vehicles (EVs). These types of batteries can also store significant amounts of energy from solar and wind power generators, thus leading to a fossil fuel-free society.

Electronics with lithium-ion battery

Lithium-ion battery, having much better energy density (joules of energy per kilogram) than previous battery technologies like nickel-cadmium (NiCd) and nickel-metal-hydride (NiMH), makes use of portable electronics more versatile to mankind. Lithium is not the most stable element, so many new safety concerns arose with the introduction of lithium-ion batteries.

Sophisticated protection electronic circuits have been designed and developed to make lithium-ion batteries safe to use. Lithium-ion 18650 cells, which are shaped like cylinders (with 18mm diameter and 65mm length), are commonly used in present-day laptops, whereas typical notebook computers need battery packs comprising an average of about four to six 18650 cells to allow it to operate for several hours between charges.

Then about a decade ago, lithium-polymer (lipo) cells made their entry to the consumer market. Because of their attractive size characteristics like thin, rectangular, customised pouch-like shape, these packs were very attractive to designers looking for space savings, especially in ultra-slim electronic devices. Presently, lithium-polymer cells are being used in thin and large-screen electronic devices like smartphones and slimmer notebooks combined with high-end processors to run complex programs while still providing instant-on capability.

All of these trends in consumer

electronics have driven the demand for larger, higher-capacity lithium-polymer cells with higher discharge currents. These demands and applications are fulfilled with higher current ratings in battery protection devices.

Working of a lithium-ion battery

Lithium is not only the lightest of all metals but also has the greatest electrochemical potential and, thus, provides the largest energy density by weight. Because of the inherent metallic instability of lithium, initial efforts to develop rechargeable lithium batteries failed miserably due to safety reasons.

Research then shifted towards a non-metallic lithium battery using lithium ions. Lithium-ion battery is slightly lower in energy density than lithium metal but is safe taking into consideration certain precautions while charging and discharging of the battery. As with most batteries, lithium-ion battery also has an outer case made of metal, which is particularly important here because the battery is pressurised. Metal case of the lithium-ion battery, thus, has some kind of pressure-sensitive vent hole and, if the battery ever gets so hot that it risks exploding from over-pressure, this vent releases the extra pressure acting as a safety measure. The vent acts as a positive temperature coefficient (PTC) switch keeping the battery safe from overheating. The metal case holds a long spiral comprising three thin sheets pressed together as positive electrode, negative electrode and separator, which are submerged in an organic solvent acting as the electrolyte.

Ether is one common solvent used in lithium-ion batteries. Positive and negative electrodes are separated by a very thin sheet of micro-perforated plastic, allowing ions to pass through. The positive electrode is made of lithium-cobalt-oxide (LiCoO_2), while the negative electrode is made of carbon. During the battery charging process, ions of lithium move through the electrolyte from the positive electrode to the negative electrode and attach to the carbon. Whereas, during discharging, lithium-ions move back to LiCoO_2 from carbon. Movement of lithium ions takes place at a fairly